Customer Behavior & Purchase Trend Analysis

Understanding customer behavior through data to drive strategic business decisions.

Project Overview & Goals

Data Insights

Analyze 3,900 customer records to uncover buying habits and preferences.

Visual Storytelling

Create clear visuals to highlight key patterns and customer segments.

Business Impact

Improve marketing, target customers, and support better decisions.

Dataset at a Glance

Our dataset comprises **3,900 customer records** across **18 columns**.

A key finding was **37 missing values** in the **Review Rating** column, which were addressed during cleaning.

3,900

Customer Records

Extensive dataset for robust analysis.

18

Data Columns

Rich detail on customer behavior.

37

Missing Values

In Review Rating, imputed for accuracy.

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Key Data Features

Customer Demographics

Age, Gender, Location – for targeted marketing strategies.

Purchase Details

Items, Amount, Category, Season, Review Rating – for product demand and satisfaction.

Shopping Behavior

Subscription, Discounts, Frequency – for loyalty and retention strategies.

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Data Preparation & Cleaning

01

Data Loading & Exploration

Imported dataset with pandas; used `df.info()` and `.describe()` for initial checks.

02

Missing Data Handling

Imputed missing 'Review Rating' values using median per product category.

03

Column Standardization

Renamed columns to snake case for improved readability.

04

Feature Engineering

Created 'age_group' and 'purchase_frequency_days' columns.

05

Database Integration

Loaded cleaned data into PostgreSQL for SQL analysis.

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Key SQL Analysis Insights

Revenue by Gender

Male customers generated **$157,890**, Female **$75,191**.

Top 5 Products by Rating

Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78).

Shipping Type Comparison

Express shipping had a slightly higher average purchase amount (**$60.48**) than Standard (**$58.46**).

Subscribers vs. Non-Subscribers

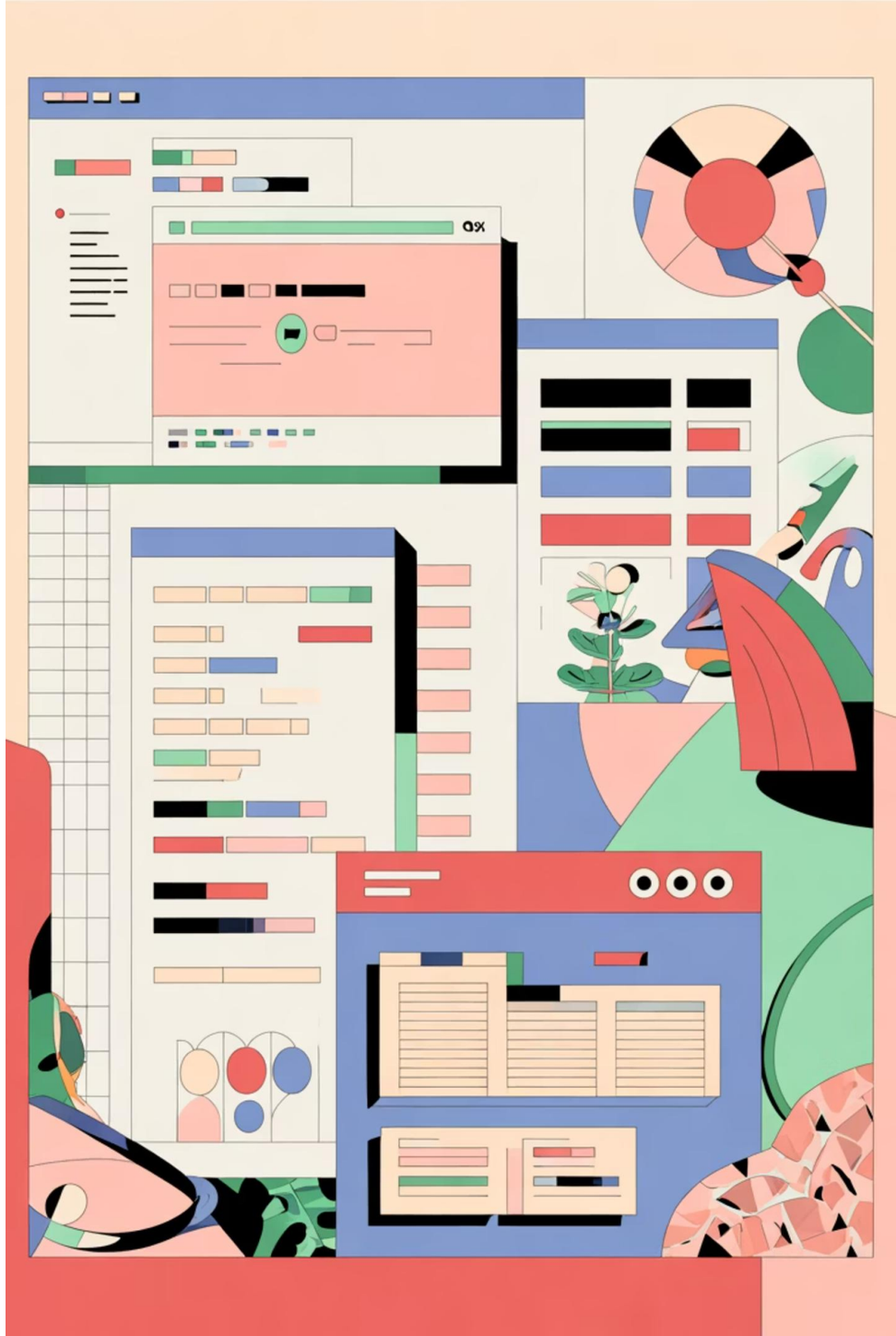
Non-subscribers contributed **$170,436** total revenue vs. subscribers' **$62,645**.

Discount-Dependent Products

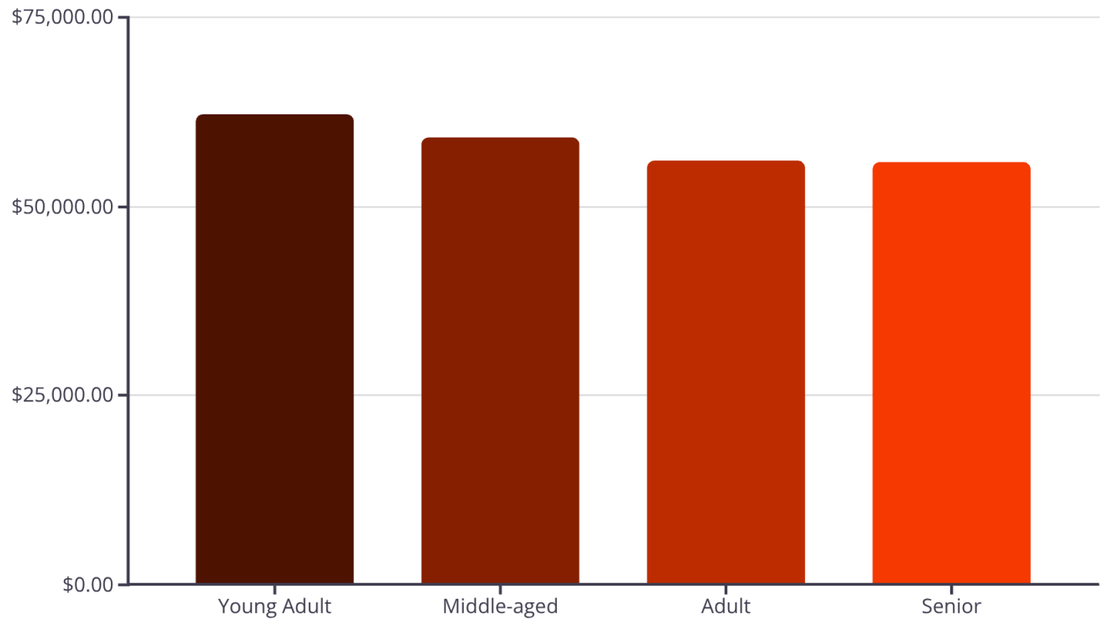
Hat (50%), Sneakers (49%), Coat (49%), Sweater (48%), Pants (47%) had highest discount usage.

Customer Segmentation

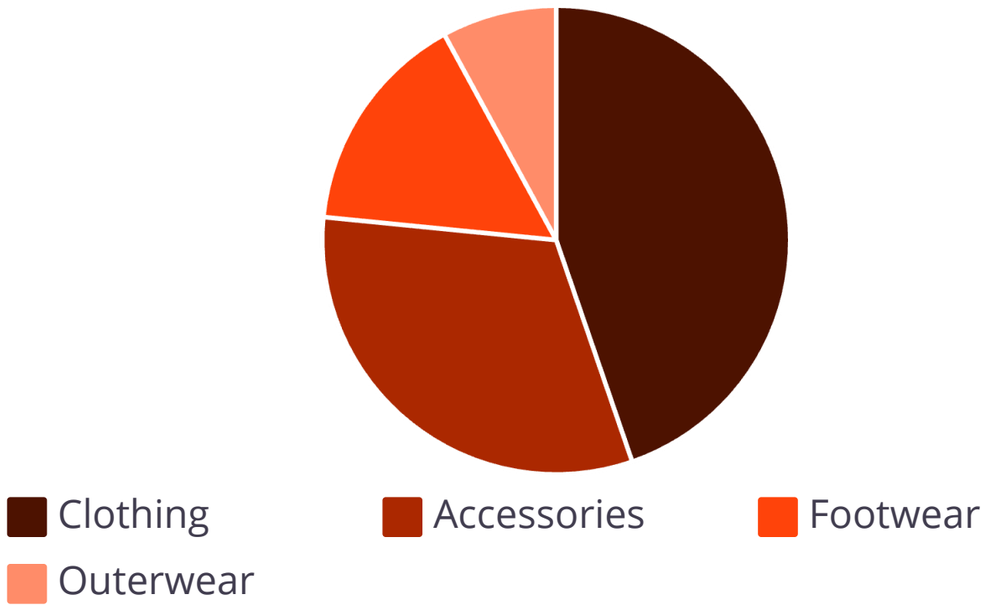
Loyal (3116), Returning (701), New (83) customers identified.



Revenue by Age Group & Category



Young Adults and Middle-aged groups drive the highest revenue.



Clothing and Accessories are the dominant revenue categories.

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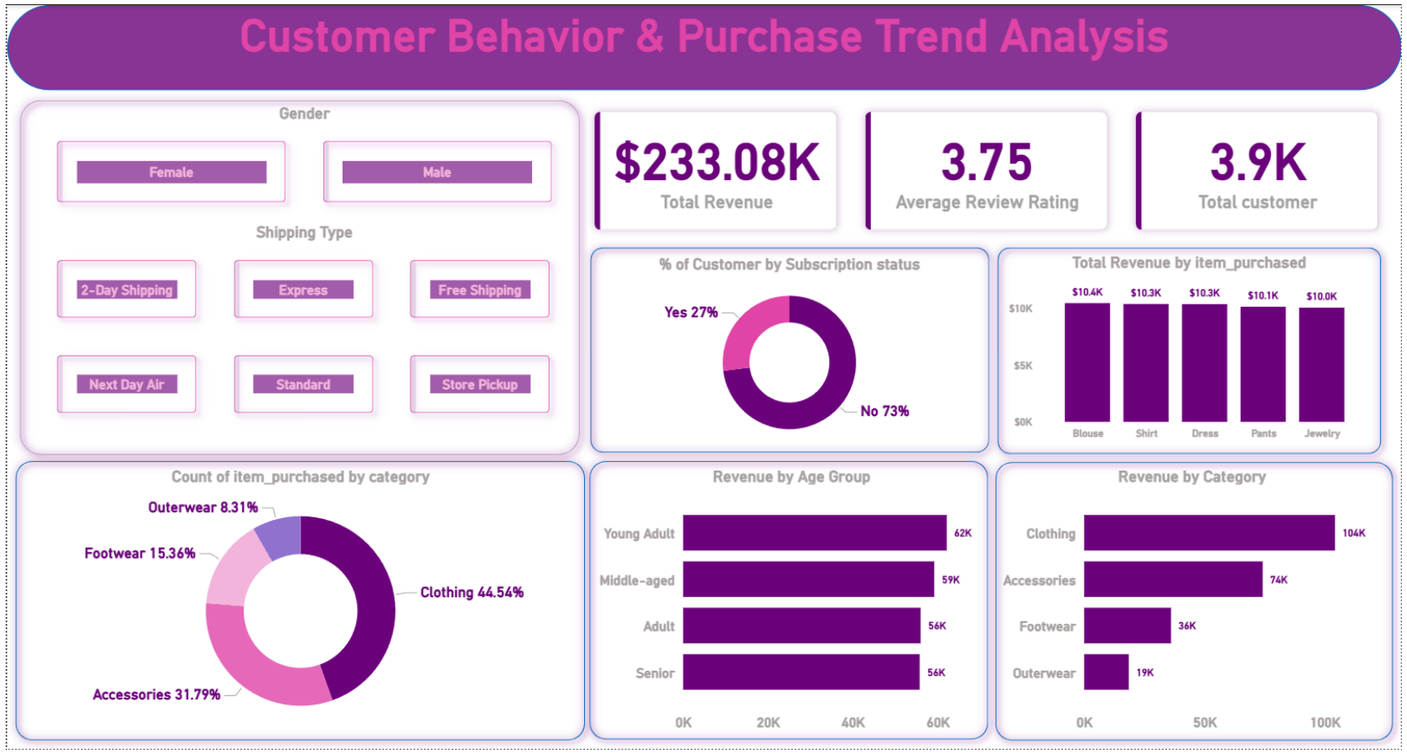
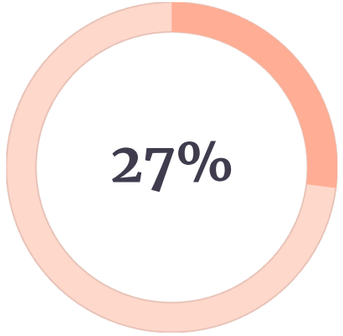
Power BI Dashboard Highlights

Interactive dashboard provides visual insights into customer behavior and sales performance.



Total Revenue

Total Customers



Subscribed Customers

Avg. Review Rating

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Strategic Business Recommendations

Boost Subscriptions

Offer exclusive perks to increase the 27% subscriber base.

Strengthen Loyalty

Implement reward points and tier levels for repeat customers.

Optimize Discounts

Analyze effectiveness and target promotions for high-value categories.

Leverage Best Sellers

Highlight top products like Blouses, Shirts, and Jewelry in campaigns.

Targeted Growth & Improvement

1

Target High-Revenue Segments

Tailor campaigns for Young Adult and Middle-aged groups.

2

Improve Shipping Strategy

Incentivize faster options to increase order value and satisfaction.

3

Expand High-Performing Categories

Increase variety in Clothing and Accessories, optimizing stock.

4

Enhance Review Engagement

Encourage feedback to refine product quality and service from 3.75 average rating.

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